

Capstone Project One-Pager

Selected Topic: Draftly: Gmail AI Reply Agent

Project Goal: Build a backend system that connects securely with Gmail, reads recent or unread emails, generates AI-powered reply drafts, and lets the user review, edit, approve, or reject every draft before anything is sent.

Step-by-Step Approach

1. Understand requirements and scope

Study the Draftly case study and identify the core flows: Gmail connection, email fetch, AI draft generation, review/approval, sending, logging, and error handling. Define the MVP as a backend-first system with REST APIs and a simple review workflow.

2. Design the system architecture

Plan modules for authentication, Gmail integration, AI draft generation, draft management, sending, logging, and user preferences. Design database tables for users, OAuth tokens, fetched emails, drafts, statuses, logs, and preferences.

3. Implement Gmail authentication

Use OAuth2 for secure Gmail access, store encrypted access/refresh tokens, and support logout or token revocation.

4. Fetch and process emails

Use the Gmail API to fetch unread or recent emails with sender, subject, body, timestamp, message ID, and thread ID. Clean and structure email content before passing it to the AI model.

5. Generate AI reply drafts

Create prompts using email context, thread history, tone preference, user signature, and recent sent-email style. Generate replies in formal, concise, or friendly tones.

6. Build review and approval APIs

Provide endpoints to view, edit, approve, reject, and track drafts. Ensure only approved drafts can move to the sending stage.

7. Send replies safely

Send approved replies through the Gmail API while preserving thread metadata. Add retries, idempotency, and failure handling for expired tokens, API errors, or failed sends.

8. Test, document, and deliver

Test the end-to-end flow: connect Gmail, fetch email, generate draft, approve, send, and log result. Prepare README documentation, API details, design decisions, GitHub repository link, and demo video.

Expected Outcome

A reliable, secure, and extensible Gmail AI assistant backend that reduces repetitive email drafting while keeping the user fully in control.